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Fracture Surface Energy of Glass

S. M. WIEDERHORN

National Bureau of Standards, Washington, D. C. 20234

Fracture surface energies of six glasses were measured using the double-cantilever cleavage technique. Values ranged from 3.5 to 5.3 J/m² depending on the chemical composition of the glass and the temperature of the test. The fracture surface energy increased with decreasing temperature and increasing Young's modulus; however, exceptions to this behavior were noted. The magnitude of the values obtained is discussed with respect to the theoretical strength of glass and possible irreversible effects at the crack tip such as stress corrosion and plastic deformation are considered.

I. Introduction

Crack propagation is characterized by the conversion of elastic strain energy into surface energy. Crack motion can occur only if the elastic strain energy decrease during fracture exceeds the surface energy increase. The energy associated with surface formation during fracture has been termed the fracture surface energy. It controls the fracture process and is important in understanding the underlying mechanisms affecting the fracture of solids.

Griffith[1] was the first to recognize the importance of surface energy to the fracture process and formalized this recognition into a concrete fracture criterion. For an elliptically shaped crack in a two-dimensional medium, he showed that crack propagation occurs when the following formula is satisfied:

$$S = \sqrt{2E\gamma/\pi L} \tag{1}$$

where S is the stress applied to the specimen, E Young's modulus, L the half-length of the internal crack, and γ the surface energy. Formulas of this type hold for a large range of crack and specimen geometries.[2] The fracture load always depends on the square root of the surface energy multiplied by a function of the elastic constants and modified by some function of the crack and specimen dimensions and shape. Thus a determination of the fracture surface energy of a material allows in principle a determination of the strength of a body for any random crack shape or specimen geometry.

The fracture surface energy defined in Eq. (1) is the energy associated with the formation of a unit area of the surface formed by the fracture process. It represents the work per unit area of crack surface and usually contains contributions that reflect nonconservative processes at the crack tip such as plastic deformation and irreversible chemical reactions. Because of these irreversibilities, the fracture surface energy generally is not equal to the specific surface free energy of the surfaces created during the fracture process. The latter term is the reversible work per unit area of surface formed.[3,4]

Glass is an excellent material on which to measure fracture surface energies. Linear elastic theory is applicable; glass is isotropic so that fracture surface energies do not depend on crack plane orientation in the material; and it can be made homogeneous so there is no concern with the interaction of the crack tip with internal flaws or grain boundaries. Finally, glass is one of the most brittle materials known, so that plastic deformation at crack tips is minimal. The one limitation of glass is its extreme susceptibility to stress corrosion by water vapor in the atmosphere.

The purpose of this paper is to present and discuss some results of a study on the fracture of glass. Experimental frac-

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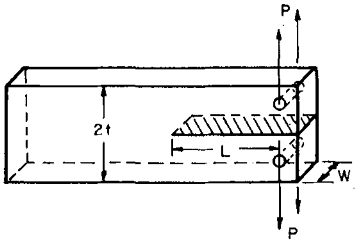

Fig. 1. Specimen configuration. Cross-hatched area designates fracture surface; direction of propagation is from right to left.

ture surface energy values were obtained on six glasses by the double-cantilever technique in dry, chemically inert environments at three temperatures. Experimental values will be discussed with respect to irreversible effects such as stress corrosion and plastic deformation.

II. Experimental Technique

The technique is essentially that described by Gilman.[5] The specimen shape is given in Fig. 1. The essential dimensions are crack length L, specimen thickness w, and height of the specimen $2t$. The experiment is conducted by increasing the applied load P until fracture results. The fracture surface energy, γ, can then be calculated from the formula

$$\gamma = 6P^2L^2/Ew^2t^3\ [1 + 1.34t/L + 0.45(t/L)^2] \qquad (2)$$

where E is Young's modulus. The experimental data can also be used to calculate critical stress intensity factors, which are also important fracture parameters.[2] The critical stress intensity factor, K_{IC}, is given by the formula

$$K_{IC} = PL/wt^{1.5}\ (3.47 + 2.32t/L) \qquad (3)$$

Equations (1) and (2) were obtained from an elastic analysis of the double-cantilever configuration. Derivation of these equations may be found in Refs. 6 and 7.

The specimens were glass microscope slides 75 by 25 by 1.5 mm. The compositions of the glasses (Table I) were selected to give a wide range of elastic moduli, which theoretically should be linearly related to the measured fracture surface energy.[8] With the exception of fused silica, all specimens were annealed before use at the manufacturer's suggested anneal point (Table I) for 30 min. They were then cooled at 2°C/h until they were at least 200° below the anneal point. The annealing was done to produce identical thermal histories and structure for all specimens of a given composition. The Young's modulus of each of the glasses was determined on ten pieces from each lot by the dynamic method described in Ref. 9.

Holes were drilled into the ends of the slides with a $^1/_{16}$-in. diamond drill. Cracks of the desired length were introduced by scribing a scratch along the midplane of the slide with a diamond stylus. The crack was induced to propagate along the scratch by applying a point force over the scratch on the face of the slide opposite the scratch. The crack was straightened and extended slightly in the testing machine by applying a light load to the ends of the slide. The crack surfaces so formed were perpendicular to the surfaces of the slide, and the leading edge of the crack was at an angle varying between 60° and 90° to the faces of the slide. To check the effect of the scratch on the fracture surface energy, measurements were made with the crack uninfluenced by the scratch. The procedure was identical to that just described except that the scratch extended only part way across the midplane of the slide, and the crack was propagated beyond the tip of the scratch. Relative crack lengths, L/t, were greater than 1.5. Before testing, all slides were heated for 1 h in dry nitrogen containing less than 0.001 mol% H_2O at approximately 300°C. This final heat treatment relieved residual stresses along the scratch and removed surface adsorbed water before the test in a dry gaseous environment.

Specimens were mounted in a universal testing machine using two small hooks which imposed little constraint on the sample. There was, nevertheless, a small unavoidable torsion which caused the crack front to lie at a 60° to 90° angle to the slide face. Crack lengths were measured to 0.01 cm using a 20 power microscope attached to a cathetometer. Other dimensions of the specimens were obtained with a micrometer and vernier calipers. The height and length of the slides were measured to 0.01 cm and the width, w, was measured to 0.003 cm.

Measurements were conducted in three environments: liquid nitrogen, 77°K; a mixture of liquid toluene and solid carbon dioxide, 195°K; and dry gaseous nitrogen, 300°K. The nitrogen gas used was super-dry or ultra-high-purity grade and had a dew point below −60°C. As reported previously,[10] a small amount of crack motion was always observed before catastrophic failure. To limit the effects of this motion, the specimen was loaded rapidly, so that the critical load was reached before the crack moved very far. Consistent results were obtained for loading rates that varied between 0.1 and 1 min total time to failure. Specimens tested in liquid nitrogen or in toluene were immersed in the liquids during the test. Specimens tested in dry nitrogen were dried and tested in the same chamber so that they never came in contact with the air during the test. The atmosphere in the test chamber was maintained by a continuous flow of dry nitrogen gas.

III. Experimental Results

Results are given in Tables II and III. Reproducibility of the technique was quite good. Standard deviations ranged from 2 to 7.5% for the fracture surface energies and from 1 to 3.5% for the critical stress intensity factors. Duplicate runs gave excellent agreement with the exception of borosilicate glass in dry nitrogen. From Table II it can be concluded that the scratch had no significant effect on the measured fracture surface energy values.

The fracture surface energy of the lead, alumina, and soda-lime glass increased approximately 15% as the temperature decreased from 300° to 77°K. The fracture energy of the borosilicate glass was approximately the same at 300°K and at 77°K, while that of silica and Vycor glass increased

Table I. Glasses Studied

Glass	Composition (wt fraction)								Young's modulus	Anneal point
	SiO_2	B_2O_3	Al_2O_3	Na_2O	K_2O	MgO	CaO	PbO	($\times 10^{10}$N/m^2)	(°C)
Silica	0.998								7.21	
96% silica	0.96	0.03	0.01						6.59	910
Aluminosilicate	.57	.04	.20	0.01		0.12	0.06		8.91	715
Borosilicate	.80	.14	.02	.04					6.37	565
Soda-lime	.72		.02	.14	0.01	.04	.07		7.34	528
Lead-alkali	.60		.04	.10	.02			0.24	6.53	470

Table II. Fracture Surface Energy of Glass (J/m^2)

Glass	$N_2(l)$, 77°K	Toluene-$CO_2(s)$, 196°K	$N_2(g)$, 300°K
Silica	4.56 ± 0.18(12*)	4.74 ± 0.35(9) 4.88 ± 0.22(10) 4.86 ± 0.35(12*)	4.42 ± 0.25(10) 4.32 ± 0.22(13*)
96% silica	4.17 ± 0.09(8)	4.60 ± 0.24(11)	3.96 ± 0.13(11)
Lead-alkali	4.11 ± 0.33(11)		3.50 ± 0.12(9) 3.55 ± 0.12(12)
Soda-lime	4.52 ± 0.19(9) 4.59 ± 0.22(10)	4.44 ± 0.18(9) 4.52 ± 0.19(8)	3.82 ± 0.10(14*) 3.91 ± 0.12(7)
Aluminosilicate	5.21 ± 0.12(10)		4.63 ± 0.23(8) 4.68 ± 0.22(14*)
Borosilicate	4.70 ± 0.24(10)		4.51 ± 0.11(6) 4.75 ± 0.13(8)

NOTE: Bracketed numbers give the number of samples used to obtain the average. Asterisks indicate the crack was extended beyond the scratch. Uncertainties are the standard deviation for each determination.

Table III. Critical Stress Intensity Factors ($\times 10^5 N/m^{3/2}$)

Glass	$N_2(l)$, 77°K	Toluene-$CO_2(s)$, 196°K	$N_2(g)$, 300°K
Silica	8.11 ± 0.16(12*)	8.33 ± 0.31(9) 8.46 ± 0.19(10) 8.37 ± 0.30(12*)	7.98 ± 0.23(10) 7.89 ± 0.20(13*)
96% silica	7.41 ± 0.07(8)	7.79 ± 0.20(11)	7.22 ± 0.12(11)
Lead-alkali	7.34 ± 0.29(11)		6.77 ± 0.12(9) 6.82 ± 0.11(12)
Soda-lime	8.17 ± 0.17(9) 8.22 ± 0.20(10)	8.08 ± 0.16(9) 8.15 ± 0.17(8)	7.49 ± 0.11(14*) 7.58 ± 0.12(7)
Aluminosilicate	9.63 ± 0.11(10)		9.08 ± 0.23(8) 9.13 ± 0.22(14*)
Borosilicate	7.74 ± 0.20(10)		7.58 ± 0.10(6) 7.78 ± 0.11(8)

NOTE: Bracketed numbers give the number of samples used to obtain each average. Asterisks indicate the crack was extended beyond the scratch. Uncertainties are the standard deviation for each determination.

between 300° and 196°K and then decreased between 196° and 77°K.

In theory, the critical stress intensity factor of a material is proportional to the strength of that material[2] and may be used as an engineering parameter to evaluate the relative strength of different materials. If the relation between the critical stress intensity factor and strength holds for the glasses studied, it follows from Table III that the high-alumina glass is the strongest of the 6 tested, being approximately 30% stronger than the high-lead glass, which was the weakest one tested. This conclusion should be valid in the absence of stress corrosion effects.

Fracture surface energies from Table II were plotted as a function of the Young's modulus of the glass in Figs. 2 and 3. At liquid nitrogen temperature (Fig. 2) the fracture surface energy seems to be a linear function of the Young's modulus for 5 of the glasses used, the only exception being the borosilicate glass. At room temperature (Fig. 3) the data scatter

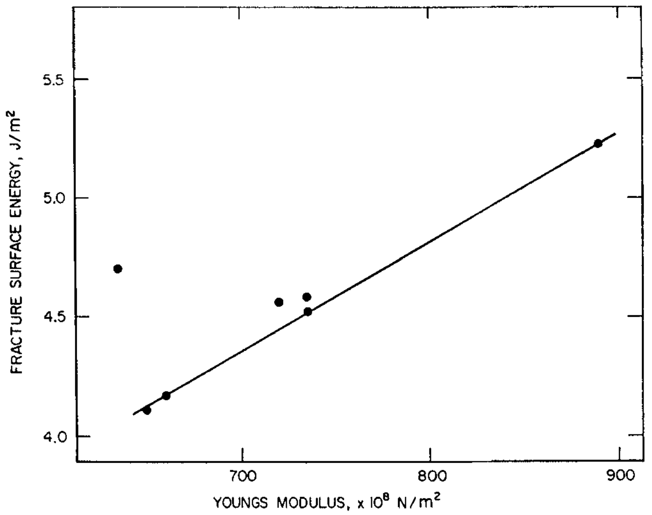

Fig. 2. Dependence of fracture surface energy on Young's modulus, 77°K (liquid nitrogen).

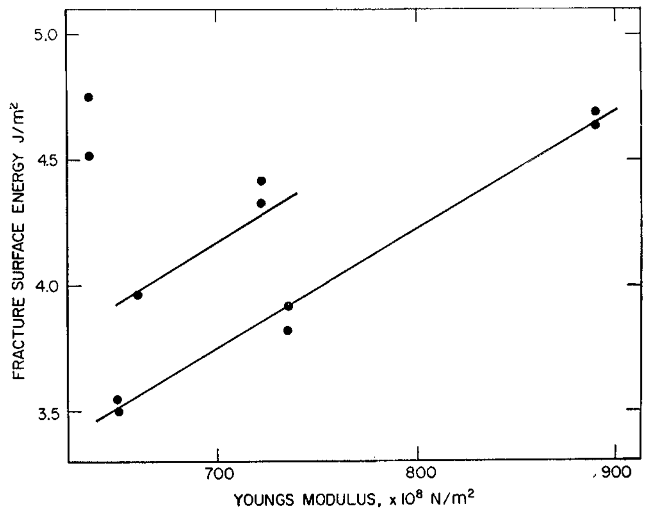

Fig. 3. Dependence of fracture surface energy on Young's modulus, 298°K (dry gaseous nitrogen).

Table IV. Fracture Surface Energies of Glass, Other Studies

Glass	Fracture surface energy (J/m²)	Ref.
Soda-lime	4.06	11, in vacuum
"	1.8–10	12
"	4–7	13, work-of-fracture method
"	3.4–5.2	14, work-of-fracture method
"	6–7	13
"	8–11	13, compliance method
Lead-alkali	1.16	15
Soda-lime	1.70	15
96% silica	3.48	15
Aluminosilicate	3.34	15

NOTE: All measurements, except where stated, were obtained by the analytical technique which uses a Griffith-type equation to relate the fracture surface energy to the breaking load. The compliance method and the work-of-fracture method are described in Refs. 2 and 13 and 14, respectively. All measurements were made at room temperature.

more widely and cannot be described by a single straight line. The lead, alumina, and soda-lime glass data did, however, lie on a line, and the fused silica and Vycor glass were both displaced about the same distance from this line. The linear behavior of the fracture surface energies in Fig. 2 is in agreement with the surface energy expression of Gilman,[8] who showed that the surface energy of a solid should be linearly related to its Young's modulus. The scatter in Fig. 3 and the deviation of the borosilicate glass from the curve in Fig. 2 show that this relation is only approximate for the glasses studied.

IV. Discussion of Results

(1) *Fracture Surface Energy Values*

The data of Table II are compared with values from other studies in Table IV. The results most comparable to those of this study are those by Berdennikov,[11] who measured the effects of several environments on the fracture surface energy of soda-lime glass using a center slot configuration for which the fracture surface energy can be calculated from Eq. (1). Unfortunately, Berdennikov did not use this equation in the original work and, as a result, the fracture energy values calculated by him were low. These values were corrected in Ref. 10. The fracture surface energy value presented in Table IV was obtained by Berdennikov in vacuum and is in excellent agreement with the soda-lime glass value in Table II. It is reassuring that the same fracture energy is obtained for soda-lime glass even though the specimen configuration is somewhat different.

Roesler[12] measured the fracture surface energy of glass using a flat-ended punch that induced a conically shaped crack to grow from the edge of the punch into the glass. An approximate analysis was then used to estimate the fracture surface energy, and the values obtained depended on the load duration, ranging from 10 J/m² for a load duration of a few seconds to 1.8 J/m² for loading over long periods. The high values were attributed to anelastic effects at the crack tip, whereas the low values were attributed to fracture surface energy lowering caused by chemical changes and adsorption of vapors at the crack tip. Roesler selected a 15 min value of the fracture, 4.1 J/m², as a good approximation of the true fracture energy of glass. This value is in excellent agreement with those obtained in this work. It should be noted that the method used by Roesler propagated the crack under combined shear and tensile loading, whereas the other experimental determinations reviewed in Table IV were obtained by propagation of cracks under tensile loading. This difference in loading technique might be expected to lead to different values of the fracture surface energy under some physical circumstances; i.e. if plastic deformation were occurring at the crack tip. Although the values of Roesler are of the same order of magnitude as those obtained in this work, a more quantitative comparison of the data is not possible because of the wide range of the values obtained by Roesler.

Fracture surface energies were obtained for glass by Davidge and Tappin[13] and by Nakayama[14] by the work-of-fracture technique, in which the actual work to propagate a crack over a complete cross section is determined. Values presented in Table IV were obtained in air on moving cracks, and it is believed that the motion was sufficiently rapid to exclude the effect of moisture. Results obtained by Davidge and Tappin and by Nakayama are in the same range as those of this study. Note that the work-of-fracture method gives the surface energy required for continuing crack motion rather than that required to initiate crack motion. In principle, these two values should be the same for completely brittle materials. A further discussion of this point is given in Ref. 13.

Davidge and Tappin[13] also determined the fracture surface energy of glass by the compliance method and the analytical method, in which a Griffith-type equation is used to determine the fracture surface energy from experimental data. Specimens were loaded in three-point loading using edge-cracked bend specimens. Experimental values were higher than those obtained in this paper (Table IV). The difference may arise from the use of notches rather than sharp cracks to initiate crack motion.

The fracture surface energies of several glasses of compositions identical to those used in this work were determined by Shand,[15] who introduced cracks of known dimension into the face of glass sheets and then broke the sheets in flexure. Surface energies were calculated from a Griffith-type formula and are presented in Table IV. Values are lower than those obtained in the present work. The difference undoubtedly arises from the fact that the runs were conducted in air containing moisture, which is known to reduce the force required for catastrophic failure of glass.[16] An idea of the relative sensitivity of the various glasses to attack by moisture can be obtained from Shand's data. The lowering of the fracture surface energy is seen to be greater for the high-lead and the soda-lime glass, indicating that these glasses are more sensitive to the presence of water vapor than the other glasses tested.

A theoretical estimate of 1.75 J/m² for the surface energy of silica glass was obtained by Charles,[16] who assumed the surface energy to be equal to the sublimation energy of the material associated with the surface. This value is about one-fourth of the values reported in this paper. Despite the crudeness of the approximation of the theoretical estimate, the values obtained in the present work do seem high. Part of the difference might be due to nonconservative effects at the crack tip, such as plastic deformation.

At high temperatures, glass becomes fluid, and the surface energy can be determined by procedures usually reserved for liquids.[17] Experiments of this type indicate that the surface energy of glass increases with decreasing temperature in an approximately linear manner.[18] By extrapolating the data to room temperature,[1] values of approximately 0.5 J/m² are obtained. This value is about 10% of the surface energy values determined by fracture techniques in this and other experiments. As suggested by Morey,[18] identical values would not be expected. At high temperatures, the components of the glass are mobile and are able to attain an equilibrium structure which reduces the surface free energy of the glass to a minimum. At low temperatures, the glass components are not mobile and cannot rearrange themselves. Therefore, fracture surface energies reflect a nonequilibrium configuration of the glass surface and are greater than surface energies obtained by extrapolation from high temperatures.

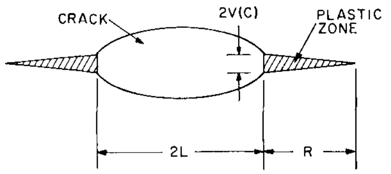

Fig. 4. Dugdale model of plastic deformation near a crack tip. After Hahn and Rosenfield, Ref. 37.

(2) *Fracture Reversibility*

Fracture as originally envisioned by Griffith[1] was a reversible process. Cracks could be propagated or healed by increasing or decreasing the load applied to the cracked body, and the energy absorbed by the surface during the fracture process would be released during the healing process. Reversible fracture is seldom observed, although crack healing has been reported in mica[19-21] and alkali halide single crystals.[22-24] Most often cracks that have propagated part way through a body remain after the constraints have been released. The fact that cracks remain open is often given as evidence that fracture is an irreversible process.

Two important processes that affect the reversibility of fracture are plastic deformation and stress corrosion at the crack tip during fracture. Both processes prevent rehealing, the first by displacing material during failure and the second by removing material and by chemical change. Plastic deformation generally increases the measured fracture energy because elastic energy is required for the plastic deformation. Stress corrosion decreases the measured fracture energy because part of the energy required to form a new surface comes from the chemical reaction. Because these processes affect the measured fracture surface energy of solids, it is of interest to investigate the magnitude of the effect of plastic deformation and stress corrosion on the data in this report. Other processes that may affect the reversibility of crack propagation are phonon generation during crack motion and the reaction of thermal fluctuations with the crack tip to aid crack propagation. The latter process was discussed by Hillig[25] and was shown to be unimportant for large cracks. Phonon generation will not be discussed because of the lack of theory with which to make quantitative comparisons.

(*A*) *Stress Corrosion:* Stress corrosion of glass by water in the atmosphere is a well-known phenomenon that has been studied for over forty years.[16] The effect of water vapor is severe enough to reduce the strength of glass to about 30% of its water-free strength. Reductions of this magnitude correspond to 90% reductions in fracture surface energy. Thus, glass having a water-free fracture energy of 4.0 J/m^2 would have a fracture energy of 0.4 J/m^2 in the presence of water. Fracture energy reductions of this magnitude were observed by the writer on soda-lime glass. In the present work, care was taken to reduce or eliminate the effect of water vapor on the fracture surface energy by testing in dry environments and at rapid loading rates. As a result of these precautions, it is felt that the surface energy values obtained represent the water-free state of the glass.

(*B*) *Plastic Deformation:* Plasticity in glass was investigated recently by Marsh,[26-28] who demonstrated plastic flow around indentations and scratch marks and correlated this flow with an apparent yield stress in glass. The plastic flow around hardness indentations results from the very high stresses that exist around the indenter during the hardness test. Similar high stresses must exist in the vicinity of the crack tip just before fracture, and it is not unreasonable to suggest that plastic deformation might initiate from the crack tip during fracture. In fact, Marsh believes that glass behaves as a non-work-hardening solid and that failure results from plastic instability at the fracture stress.

The plastic zone size can be estimated from the Dugdale[29] model of plastic flow at a crack tip. It is assumed that the solid is a non-work-hardening solid and that the plastic zone extends a length R in front of the crack tip as depicted in Fig. 4. The material within the plastic zone (cross-hatched regions of Fig. 4) is assumed to be at the yield stress of the solid, σ_y. The length of the zone is

$$R = (\pi/8)(K_{IC}/\sigma_y)^2 \tag{4}$$

where K_{IC} is the critical stress intensity factor. When the plastic zone extends from the crack tip a distance R, the sides of the crack are displaced a distance $2V(C)$ at the crack tip, where

$$V(C) = 4\ \sigma_y\ R/\pi E \tag{5}$$

Equations (4) and (5) are valid provided the breaking stress is much less than the general yield stress of the material, a condition that holds for glass. Equations similar to (4) and (5) can be derived from other plastic flow models and give nearly identical numerical results.[30] Energy absorption according to the above treatment is qualitatively equal to $\sigma_y V(C)$.

The size of the plastic zone at the crack tip may be estimated for fused silica and for soda-lime glass, for which the necessary experimental data are available. From indentation experiments,[28] the yield stresses for soda-lime glass and fused silica at liquid nitogen temperature are 1×10^{10}N/m^2 and 1.95×10^{10}N/m^2, respectively. Data for the critical values of the stress intensity factors are given in Table III, and data for the Young's modulus are given in Table I. The lengths of the plastic zones, R, for soda-lime glass and silica glass are calculated to be 2.6×10^{-9}m and 6.4×10^{-10}m, while the displacements, $V(C)$, are calculated to be 4.5×10^{-10}m and 2.2×10^{-10}m, respectively. These dimensions are not very much larger than those characteristic of silica-based glasses, such as the silica-oxygen distance, 1.6×10^{-10}m, and the oxygen-oxygen distance, 2.6×10^{-10}m, and consequently the amount of plastic deformation that initiates from the crack tip of these glasses is small. The very small values of the calculated plastic zone sizes dramatize the reason for the very brittle nature of glass. In the absence of large amounts of plastic flow, energy absorption during crack motion in glass, 4.5 J/m^2 for soda-lime glass and 4.3 J/m^2 for silica glass, is small compared to that obtained on metals, 10^4 J/m^2, and as a result, a crack once started tends to propagate catastrophically.

The energy absorption due to plastic flow is almost exactly equal to the measured fracture surface energies of these glasses. This conclusion should be accepted with caution since the calculations presented above are based on the assumptions of continuum mechanics and are valid if the particulate nature of matter can be neglected with respect to the size of the region under study. In the present case, the structure of the solid cannot be neglected because the calculated size of the plastic zone is of the same order of magnitude as the interatomic spacing within the solid. Therefore, the results calculated from Eqs. (4) and (5) should be considered only qualitatively correct in that small plastic zones and correspondingly small energy absorptions occur at the crack tip due to plastic deformation. Further generalization on the basis of these equations is not justified.

It is of interest to compare the crack tip displacement calculated from plastic theory with that calculated from elastic theory. The elastic displacements may be calculated using the theory of Elliot,[31] which assumes that the crack is bounded by atoms centered on planes half a lattice spacing from the fracture plane. The displacement and the stresses across the crack plane are assumed to be given by linear elastic theory. For a lattice spacing of 1.6×10^{-10}m, the atoms at the crack tip are approximately 0.7×10^{-10}m from their equilibrium positions just before fracture, which leads to a crack tip displacement, $V(C)$, of 0.7×10^{-10}m. Thus the calculated elastic displacement is smaller than the calculated plastic displacement for silica and soda-lime glasses. These displacements are equivalent to a strain of approximately 40% at fracture, which is slightly high and may suggest some form of anelastic behavior, or plastic flow, at the crack tip.

(3) *Cohesive Strength*

Despite the fact that the fracture of glass may be irreversible, the fracture surface energies presented in Table II are probably the correct ones to use in estimating the strength of glass, because irreversible effects occurring at the crack tip should be included in the surface energy term of the Griffith equation. It is of interest to use the values of Tables II and III to estimate the theoretical cohesive strength of glass. This will be done for the fused silica glass, using two approaches, one suggested by Gilman[32] in which a potential function is used to calculate the theoretical strength, and one based on the work of Elliot[31] in which a relation between the critical stress intensity factor and the cohesive strength is given. These results will be compared with experimental values of strength and one other theoretical estimate.

The method of estimating the theoretical cohesive strength suggested by Gilman[32] takes cognizance of the atomic forces holding solids together by assuming an attractive cohesive force between atom sheets bordering the fracture plane. For fracture to occur, these forces must be exceeded, and the maximum force gives the cohesive strength. The attractive force between the incipient fracture surfaces is represented by a potential function, the characteristics of which are typical of the solid under study. A good representative function for glass, which is largely covalent, is the Morse-type potential:

$$U = U_0\{\exp[-2a(x - x_0)] - 2\exp[-a(x - x_0)]\} \tag{6}$$

U represents the potential energy of the fracture surfaces separated a distance x, per unit area of fracture surface, and U_0, a, and x_0 are constants that must be evaluated experimentally. The stress on these two surfaces is equal to $\partial U/\partial x$, and the cohesive strength is calculated from the fact that the stress is a maximum when $\partial^2 U/\partial x^2 = 0$. A diagram of the Morse function and its first derivative is presented in Fig. 5. The constants of Eq. (6) are evaluated from measurable material properties such as the surface energy, which determines the depth of the potential minimum, the zero stress separation distance, x_0, which determines the position of the potential minimum, and the Young's modulus, E, which determines the curvature at the potential minimum. After the necessary mathematical manipulations are performed,[32] the equation obtained for the theoretical cohesive strength is

$$\sigma_{max} = \sqrt{\gamma E/4x_0} \tag{7}$$

Substituting the appropriate experimental values from Tables I and II into Eq. (7) and assuming the equilibrium lattice spacing to be the silicon-oxygen distance,[33] 1.62×10^{-10}m, a strength of 2.2×10^{10}N/m^2 (3.2×10^6 psi) is obtained (Table V).

The above estimate for the theoretical strength of glass is supported by the calculation of Náray-Szabó and Ladik[34] who also used a Morse-type potential to calculate the maximum cohesive strength but evaluated the constants a and U_0 theoretically. The depth of the potential well U_0 was calculated by assuming that the main force to be overcome was a coulombic force between an oxygen ion and a silica complex, and the constant a was evaluated spectroscopically. The value of the theoretical cohesive strength so obtained was 2.42×10^{10}N/m^2 (3.5×10^6 psi).

In his analysis of the conditions for rupture caused by Griffith cracks, Elliot[31] discussed the stresses and displacements existing in the vicinity of a crack tip and related them to the applied load for the case of a penny-shaped crack and an elliptical crack. Expressed in terms of the critical stress

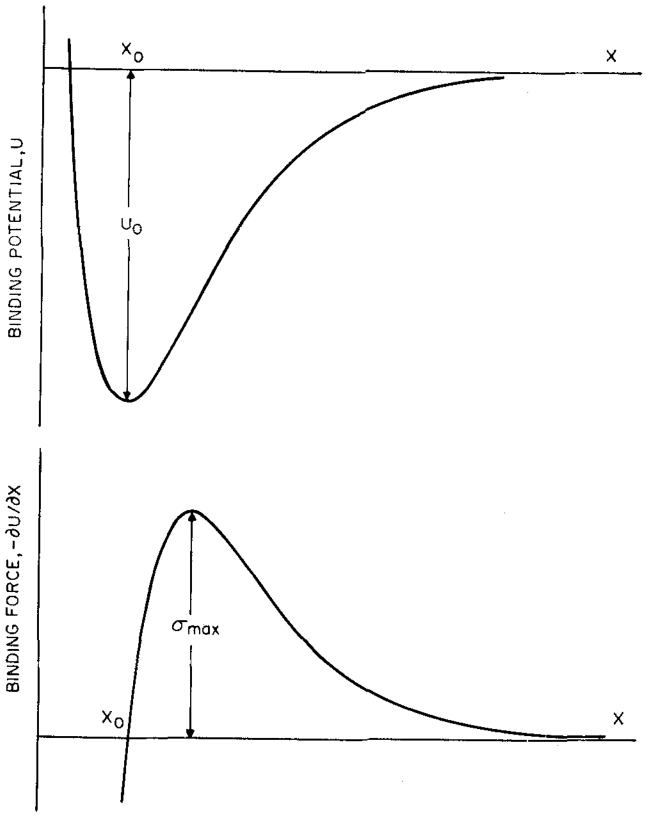

Fig. 5. Schematic of Morse potential U and its first derivative.

Table V. Comparison of Theoretical and Maximum Measured Strength of Fused Silica

Material	Strength ($\times 10^{10}$N/m^2)	Ref.
Theoretical	2.2	32
Theoretical	2.42	34
Theoretical	4.46	31
Flame-polished silica, −196°C	1.35	35
Silica fibers, −296°C	1.47	36

intensity factor, the maximum tensile stress at the tip of a crack at failure is given by the equation

$$\sigma_{max} = 0.698K_{IC}/\sqrt{x_0} \tag{8}$$

If it is assumed that crack propagation occurs when the theoretical tensile stress is exceeded at the crack tip, Eq. (8) gives the theoretical cohesive strength of the material. Substituting from Table III for K_{IC} and setting x_0 equal to 1.6×10^{-10}m, a strength of 4.46×10^{10}N/m^2 (6.5×10^6 psi) is obtained (Table V). This value is about twice that given above.

The above estimates of the theoretical strength are listed in Table V with the highest measured strengths for this material. The experimental values of strength are less than the theoretical values, ranging from 60 to 30% of the theoretical estimate. Differences between the theoretical strength and the maximum possible measured strength in a material are expected. As discussed by Hillig,[25,35] internal defects in glass such as voids and bubbles can reduce the observed strength of glass to half the theoretical limit. Even in the absence of flaws thermal fluctuations can generate flaws of sufficient severity to reduce the measured strength to half the theoretical value.[25] Considering these effects and the fact that theoretical estimates of the strength of glass are only accurate to within a factor of 2 or 3, the experimental results presented in this paper are believed to be consistent with measured strengths of glass in that they predict reasonable values for the theoretical strength.

V. Summary

The fracture surface energy and critical stress intensity factor of six glasses were measured using the double-cantilever cleavage technique. Surface energies ranged from 3.5 to 5.3 J/m^2, depending on the chemical composition of the glass and the temperature of the test. These energies compared well with published values of the fracture surface energy of glass. The fracture surface energy increased with decreasing temperature and increasing Young's modulus; however, exceptions to this behavior were noted. Cohesive strengths calculated for silica glass from the fracture energy data at 77°K ranged from 2.2×10^{10}N/m^2 (3.2×10^6 psi) to 4.46×10^{10}N/m^2 (6.5×10^6 psi) depending on the model used to calculate the cohesive strength. Measured strengths, taken from the literature, were 30 to 60% of the calculated cohesive strength. The agreement between theory and experiment is good considering the possibility of small voids or bubbles in the test specimens and the possibility of flaw generation from thermal fluctuation in the glass. Fracture surface energy values obtained are believed to be free of the effect of chemical environment but may have been affected by small amounts of plastic deformation generated at the crack tip by the high stress fields existing there just before failure.

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